



Biology 1

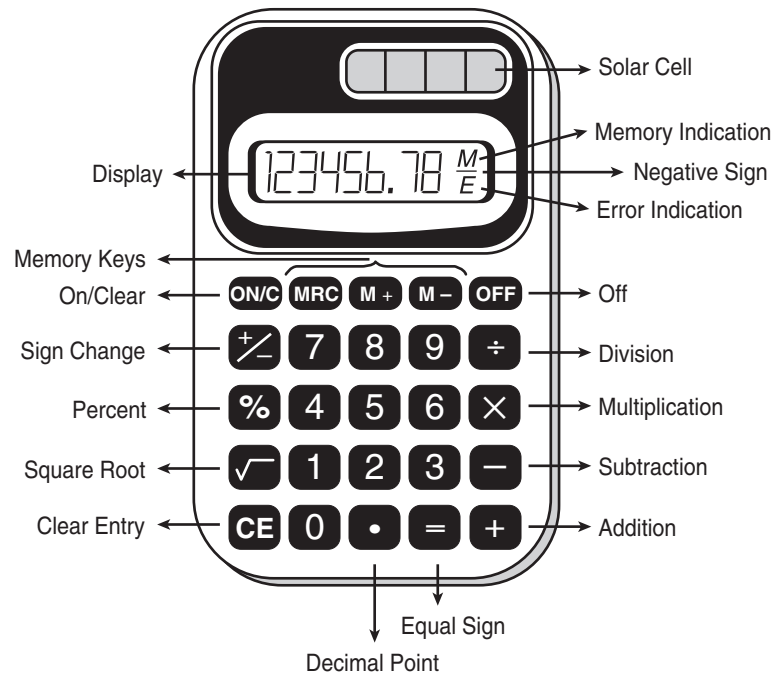
Sample Test Materials

The purpose of these sample test materials is to orient teachers and students to the types of paper-based Biology 1 End-of-Course (EOC) questions. By using these materials, students will become familiar with the types of items and response formats they may see on a paper-based test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test. The sample test materials are not intended to guide classroom instruction.

HELPFUL HINTS FOR USING A FOUR-FUNCTION CALCULATOR

This is a picture of a generic 4-function calculator and its parts.

GENERIC 4-FUNCTION CALCULATOR



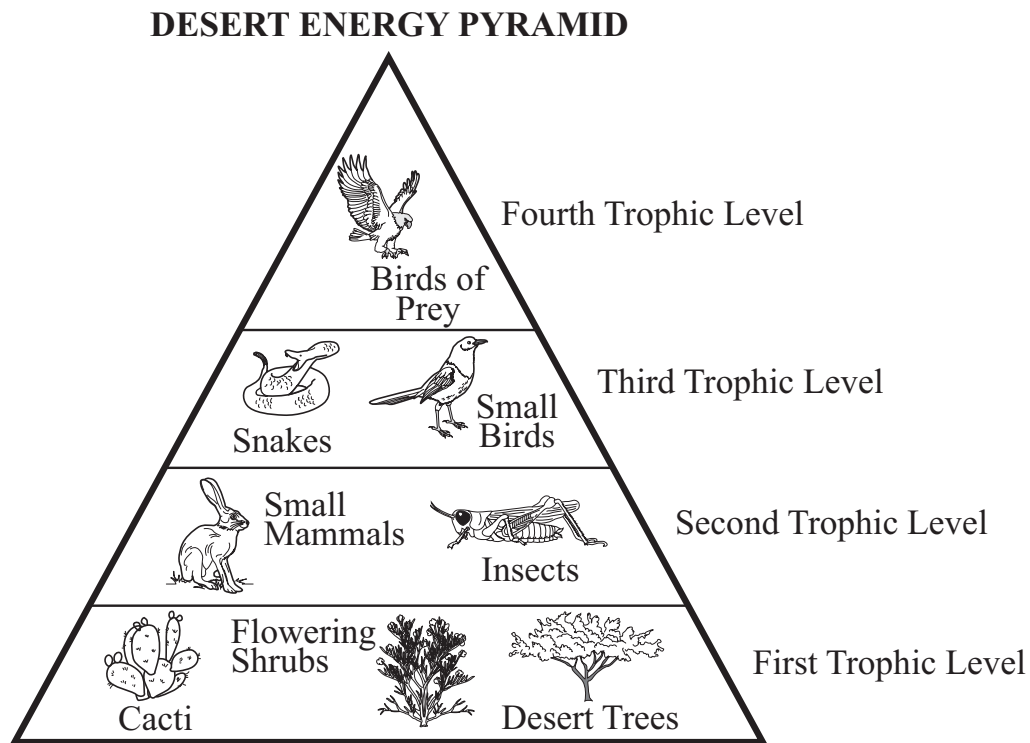
If you decide you need the calculator to help you answer a question, use the following information:

1. When starting a new problem, always clear your calculator by pressing the on/clear key.
2. If you see an **E** in the display, clear the error before you begin.
3. If you see an **M** in the display, clear the memory and the calculator before you begin.
4. If the number in the display is not one of the answer choices, check your work.
5. Remember, your calculator will NOT automatically perform the order of operations.
6. Calculators might display an incorrect answer if you press the keys too quickly. When working with calculators, use careful and deliberate keystrokes, and always remember to check your answer to make sure that it is reasonable.
7. The negative sign may appear either to the left or to the right of the number.
8. Always check your answer to make sure that you have completed all of the necessary steps.

1	2	Periodic Table of the Elements Atomic Number — 1 Element Name — Hydrogen Element Symbol — H Average Atomic Mass — 1.01 Metal Metalloid (semimetals) Non-metals										13	14	15	16	17	18
1 H Hydrogen 1.01		3 Li Lithium 6.94	4 Be Beryllium 9.01									5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
11 Na Sodium 22.99	12 Mg Magnesium 24.30											13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.95
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.87	23 V Vanadium 50.94	24 Cr Chromium 51.996	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.38	31 Ga Gallium 69.72	32 Ge Germanium 72.63	33 As Arsenic 74.92	34 Se Selenium 78.97	35 Br Bromine 79.90	36 Kr Krypton 83.80
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
55 Cs Cesium 132.91	56 Ba Barium 137.33	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.21	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
87 Fr Francium (223)	88 Ra Radium (226)	89-103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (269)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (289)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

1. A team of ecologists observed feeding patterns of several populations in the desert. The energy pyramid shown below depicts the feeding patterns the ecologists observed.



Which of the following **best** explains the difference in the amount of available energy in the trophic levels of the desert ecosystem?

- (A) There is less energy available in the producers because their tissues are less dense than those at higher trophic levels.
- (B) There is more energy available in the second trophic level because less energy is needed for hunting compared to the higher trophic levels.
- (C) There is more available energy in the birds of prey because they have greater muscle mass for storing energy than organisms in lower trophic levels have.
- (D) There is less available energy in the fourth trophic level because of the loss of energy through metabolism in each of the lower trophic levels.

- 2.** Water is essential for life. Its special properties make water the single most important molecule in plant life. Which of the following properties of water enable it to move from the roots to the leaves of plants?
- Ⓐ Water expands as it freezes.
 - Ⓑ Water is an excellent solvent.
 - Ⓒ Water exhibits cohesive behavior.
 - Ⓓ Water is able to moderate temperatures.

3. An osmosis investigation was conducted using chicken eggs to represent cells with semipermeable membranes. The mass of each egg was measured to determine how much water diffused into or out of the eggs. The eggs were first soaked in vinegar to dissolve the shell. Each egg was then placed in one of three different solutions for 24 hours. The table below shows the results of the investigation.

OSMOSIS IN CELLS

Solution	Average Mass of Eggs Before Soaking (grams)	Average Mass of Eggs After Soaking (grams)	Difference in Average Mass (grams)	Percent Change in Average Mass
Vinegar (95% water)	71.2	98.6	27.4	+38.5
Corn syrup (5% water)	98.6	64.5	34.1	-34.6
Distilled water (100% water)	64.5	105.3	40.8	+63.3

Based on this experiment, which of the following should be inferred about cells with semipermeable membranes?

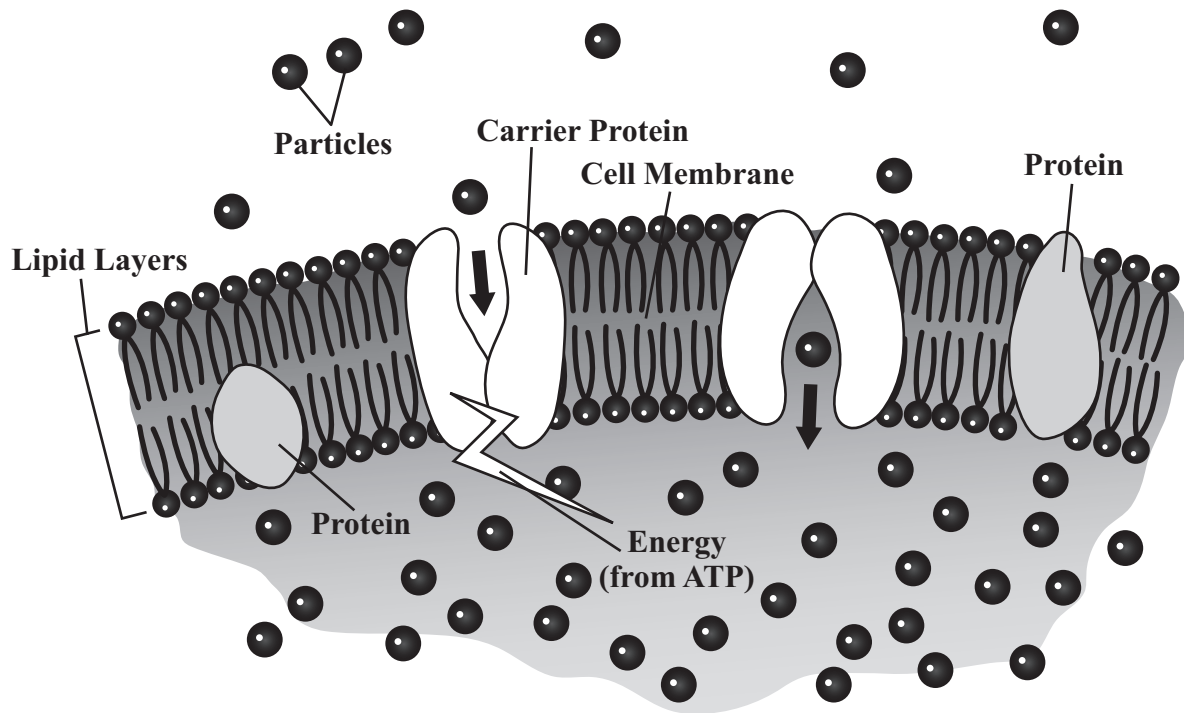
- Ⓐ Substances other than water may also cross the cell membrane.
- Ⓑ Substances other than water may block pores in the cell membrane.
- Ⓒ Water enters the cell when placed in environments of high water concentration.
- Ⓓ Water leaves the cell when placed in environments with a low concentration of solutes.

4. One of the accepted scientific theories describing the origin of life on Earth is known as chemical evolution. According to this theory, which of the following events would need to occur **first** for life to evolve?
- Ⓐ onset of photosynthesis
 - Ⓑ origin of genetic material
 - Ⓒ synthesis of organic molecules
 - Ⓓ formation of the plasma membrane

5. Hemophilia is a sex-linked, recessive trait. Which of the following describes the probability of hemophilia in the offspring of a man who does not have hemophilia and a woman who is a heterozygous carrier?
- Ⓐ There is a 0% chance that their daughters will have hemophilia.
 - Ⓑ There is a 25% chance that their sons will have hemophilia.
 - Ⓒ There is a 50% chance that their daughters will have hemophilia.
 - Ⓓ There is a 100% chance that their sons will have hemophilia.

Cell Membrane

A cell membrane is composed of a double layer of lipids in which many kinds of proteins are embedded. Many of these proteins act like gates, allowing only certain particles to enter or leave the cell.



When a particle has to go against its concentration gradient through one of these gates, energy is supplied by the cell to the correct protein to move the particle through the membrane. A sodium-potassium ion pump is an example of a type of carrier protein that uses a large portion of the cell's energy to move sodium ions through the cell membrane.

To answer this question, refer to the “Cell Membrane” passage and figure.

- 6.** The sodium-potassium ion pump found in some cell membranes is made of which of the following basic structural components?
- Ⓐ amino acids
 - Ⓑ fatty acids
 - Ⓒ monosaccharides
 - Ⓓ nucleotides

To answer this question, refer to the “Cell Membrane” passage and figure.

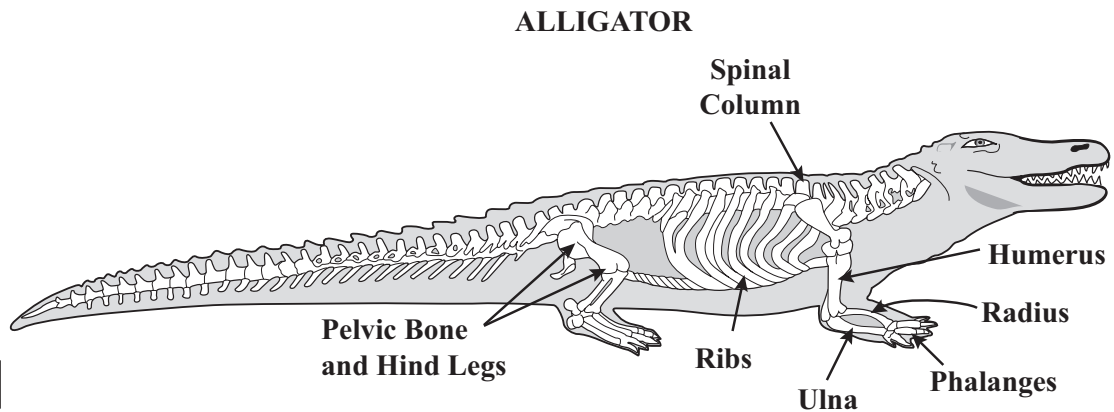
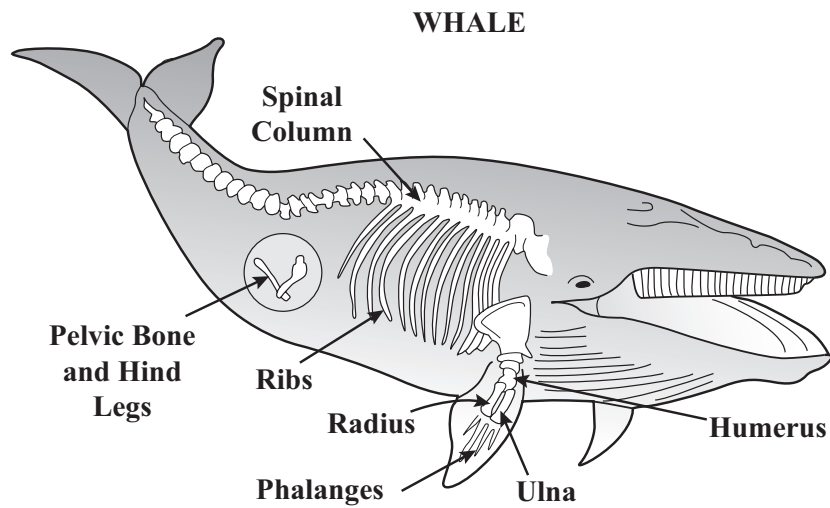
7. What is a primary function of the cell membrane?

- Ⓐ determining genetic traits
- Ⓑ defending against foreign particles
- Ⓒ breaking down proteins for energy
- Ⓓ generating energy from mineral nutrients

To answer this question, refer to the “Cell Membrane” passage and figure.

- 8.** Why does the study of cell membranes lead to a better understanding of cell function?
- Ⓐ All cell functions occur in the cell membrane.
 - Ⓑ All energy transfers occur at the cell membrane.
 - Ⓒ All cell membranes contain the information for making proteins.
 - Ⓓ All materials needed for cell functions must pass through the cell membrane.

9. The scientific theory of evolution is supported by different types of evidence. The diagrams below show the skeletons of two different animal species.

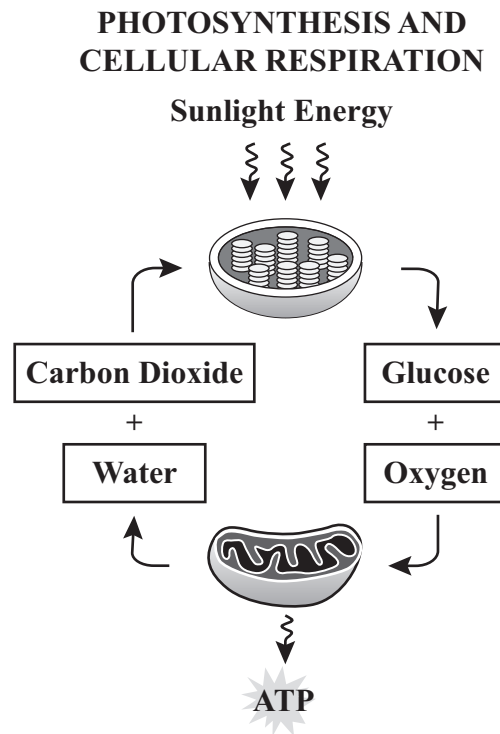


not to scale

How does comparing the skeletons of these animals provide support for the scientific theory of evolution?

- Ⓐ It provides information about the organisms' habitats.
- Ⓑ It shows possible common ancestry between organisms.
- Ⓒ It provides information to determine the organisms' ages.
- Ⓓ It shows possible chromosomal similarities between organisms.

10. The diagram below shows the relationship between photosynthesis and cellular respiration and the organelles in which they occur.



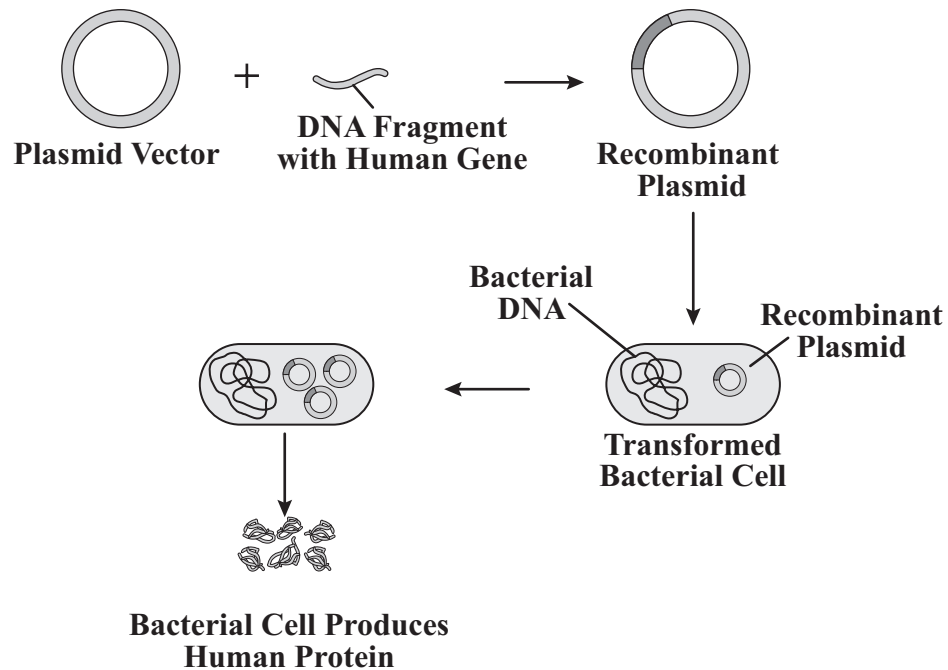
Which statement describes how photosynthesis and cellular respiration are interrelated?

- Ⓐ Oxygen is produced during cellular respiration and stored during photosynthesis.
- Ⓑ Carbon dioxide and water released by cellular respiration are used in photosynthesis.
- Ⓒ Photosynthesis releases the energy that is stored during the process of cellular respiration.
- Ⓓ Glucose is used during cellular respiration to produce food that is broken down during photosynthesis.

11. As food travels through the digestive system, it is exposed to a variety of pH levels. The stomach has a pH of 2 due to the presence of hydrochloric acid (HCl), and the small intestine has a pH ranging from 7 to 9. HCl converts pepsinogen into pepsin, an enzyme that digests proteins in the stomach. Which of the following **most likely** happens to pepsin as it enters the small intestine?

- Ⓐ It becomes inactive.
- Ⓑ It begins to replicate.
- Ⓒ Its shape changes to engulf large proteins.
- Ⓓ Its activity increases to digest more proteins.

- 12.** Genes for medically important proteins can be cloned and inserted into bacteria, as shown in the diagram below.



Why can bacteria recognize a human gene and then produce a human protein?

- Ⓐ DNA replication in bacteria and humans is the same.
- Ⓑ Bacterial cells contain the same organelles as human cells.
- Ⓒ The basic components of DNA are the same in humans and bacteria.
- Ⓓ Bacterial cells and human cells contain the same kind of chromosomes.



Biology 1, Civics, and U.S. History

End-of-Course Assessments

Answer Keys for Practice Tests

These correct answers are being provided as a reference for teachers and students who are using the EOC Practice Tests.

Biology 1 EOC Assessment Practice Test

1. D
2. C
3. C
4. C
5. A
6. A
7. B
8. D
9. B
10. B
11. A
12. C

Civics EOC Assessment Practice Test

1. C
2. D
3. C
4. A
5. D
6. A
7. C
8. D

U.S. History EOC Assessment Practice Test

1. B
2. A
3. C
4. D
5. B
6. D